

Different structured random-matrix patterns can have all the same starred moments

Run `fa_banach_001`, agent lane 10

2026-08-12

Source question

The source is Tapesh Yadav, *Wigner type laws for structured random matrices* [1], available as arXiv:2201.00257. A pattern is a set $S \subset [0, 1]^2$ with null boundary. The source fills the corresponding grid cells with centered iid entries of variance one, scaled by $N^{-1/2}$, and proves convergence of all normalized starred moments. Section 5.3 asks:

If two patterns are not similar, must their collections of limiting starred moments be different?

Here “similar” means that their approximating matrices are similar for infinitely many sizes. The answer is no.

5.3. Pattern and base change. We call a pattern S_1 to be similar to pattern S_2 if the approximating matrices for S_1 are similar to the approximating matrices for pattern S_2 infinitely often as the size of matrices increases. Since trace is basis independent, we get the same moments for similar matrices. Hence, the moments corresponding to similar patterns are the same.

A natural question is to characterize similar patterns (for example, existence of kernel functions relating two patterns) without going to the level of approximating matrices. Another question would be to understand if similar shapes produce a unique collection of starred moments. That is, if two patterns are not similar, do we always get non identical starred moments.

Result

For $S \subset [0, 1]^2$, write $K_S = \mathbf{1}_S$. Call S *p-biregular* if

$$\int_0^1 K_S(x, y) dy = p \quad \text{for a.e. } x, \quad \int_0^1 K_S(x, y) dx = p \quad \text{for a.e. } y. \quad (1)$$

Theorem 1 (Biregular moment collapse). *Let S satisfy (1). For every starred word C of length $2n$,*

$$m_C(S) = p^n m_C([0, 1]^2), \quad (2)$$

where $m_C(S)$ is the limiting normalized expected starred moment of the source. The corresponding odd and unbalanced moments vanish. Consequently, all p -biregular patterns have the same complete collection of limiting starred moments.

This gives an infinite-dimensional failure of injectivity, not merely an isolated coincidence. The following two elementary patterns make it a full counterexample to the source question:

$$S_+ := ([0, \tfrac{1}{2}]^2) \cup ([\tfrac{1}{2}, 1]^2), \quad (3)$$

$$S_- := ([0, \tfrac{1}{2}] \times [\tfrac{1}{2}, 1]) \cup ([\tfrac{1}{2}, 1] \times [0, \tfrac{1}{2}]). \quad (4)$$

Changing their boundary points is immaterial. In block notation their indicators are

$$K_{S_+} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K_{S_-} \sim \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Both are 1/2-biregular and hence have identical limiting starred moments. Nevertheless, their approximating matrices are almost surely never similar from size two onward when the allowed iid entry law is continuous.

Why all moments collapse

The proof is the tree structure hidden in the source's path-counting formula.

Lemma 1 (Oriented tree integral). *Let T be a tree with n edges, each oriented arbitrarily. If S is p -biregular, then*

$$\int_{[0,1]^{V(T)}} \prod_{u \rightarrow v \in E(T)} K_S(x_u, x_v) \prod_{u \in V(T)} dx_u = p^n.$$

Proof. Choose a leaf. If its incident edge points away from the leaf, integration in the leaf variable uses the second identity in (1); if it points toward the leaf, it uses the first. In either case this removes the leaf and contributes p . Iterate over all n edges. $\square$

Proof of Theorem 1. Expand $N^{-1} \mathbb{E} \text{Tr} C(X_{N,S}, X_{N,S}^*)$. Centering and independence leave, at leading order, pairings in which n distinct entries occur once unconjugated and once conjugated. A leading pairing has $n+1$ free index classes. Its quotient graph is connected because the cyclic trace word walks through all its vertices. It therefore has $n+1$ vertices and n edges, hence is a tree.

For a fixed leading pairing, the normalized sum over its free indices tends to the integral in Lemma 1; the orientation records which of the two occurrences is unstarred. Its value is p^n , while the same pairing has weight one for the full-square pattern. Summing over the leading pairings proves (2). If the word is odd or has unequal numbers of starred and unstarred letters, no leading conjugate pairing exists, so its limit is zero. This is also a direct specialization of the source's polar constraint-path formula: a constraint path of length $2n$ on $n+1$ colors has precisely such a tree as its underlying graph. $\square$

For (3)–(4), Lemma 1 has a discrete interpretation. Every tree has exactly two two-colorings in which adjacent labels are equal (the two constant labelings), and exactly two in which they are unequal (the two bipartite colorings). Since there are 2^{n+1} block assignments, either pattern gives each leading tree the same weight $2/2^{n+1} = 2^{-n}$.

Why the two patterns are not similar

The source declares matrix entry (i, j) active when the cell with horizontal index j and vertically reversed row index i meets S in nonempty interior. This vertical reversal interchanges the two block pictures above: $S_{\neq}$ yields diagonal-block support, while $S_{=}$ yields off-diagonal- block support.

Every diagonal entry of $X_{N,S_{\neq}}$ is therefore active. No diagonal entry of $X_{N,S_{=}}$ is active for even N . For odd N , only the unique central grid cell that straddles $1/2$ is active. Choose standard complex Gaussian entries, which satisfy all assumptions of the source. For each $N \geq 2$, the two traces are different linear combinations of continuous independent variables; their equality has probability zero. Taking the countable intersection over N , almost surely

$$\mathrm{Tr} X_{N,S_{=}} \neq \mathrm{Tr} X_{N,S_{\neq}} \quad (N \geq 2).$$

Similar matrices have equal trace. Hence the two approximating matrices are not similar for any $N \geq 2$, almost surely, and the patterns are not similar in the source’s sense. Combined with Theorem 1, this proves the claimed full counterexample.

Verification and scope

The accompanying script exhaustively enumerates complex-Gaussian leading pairings for every balanced word through length ten. It checked 350 words and 1618 leading pairings; in every term both block patterns have weight 2^{-n} . This is a finite audit of the combinatorics, while the leaf argument above proves all orders.

Searches on 2026-08-12 for the exact title and exact moment-uniqueness wording, as well as searches for biregular/doubly-stochastic variance profiles and starred moments, found literature on circular laws for variance profiles but no later paper answering this pattern-injectivity question. The run’s four cheap indexes likewise contained no result for arXiv:2201.00257. This is a bounded novelty check rather than a claim to have exhausted non-arXiv sources.

The theorem fully answers the yes/no injectivity question negatively and identifies a large moment-equivalence class. It does not classify every pair of patterns with the same starred moments. Note also that the proof of non-similarity uses only the necessary trace invariant, so it does not depend on any convention about whether ordinary or unitary similarity was intended.

Human review recommendation

Review as a likely-valid full counterexample. The two useful independent checks are: (i) every leading pairing quotient really is a tree, and (ii) the vertical row-index reversal makes $S_{\neq}$, rather than $S_{=}$, the diagonal-block approximant. Neither point affects the equality-of-moments proof, but both make the non-similarity witness transparent.

References

- [1] Tapesh Yadav, *Wigner type laws for structured random matrices*, arXiv:2201.00257, 2022.